WORKED SOLUTION

STEP III 2013, Question 12

Probability and Statistics • Indicator Random Variables

STEP III 2013, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q12

The key idea

Each X_k equals 1 exactly when a new block of As starts at position k, so S counts the blocks (runs) of As in the row. The variables are indicators, so every expectation is just the probability that a few named positions hold named letters, and that depends only on how many positions are involved, not on where they are. For the variance we expand S^2 and sort the products X_iX_j into types: equal indices, adjacent indices (which turn out to be impossible) and the rest.

Solution

Throughout, all $\binom{n}{a}$ orderings are equally likely. A useful consequence: for any m *distinct* positions, the probability that they hold a prescribed pattern containing p As and q Bs ($p+q=m$) is obtained by filling the positions one at a time, for example

$$\Pr(\text{positions } r,s \text{ hold } B,A) = \frac{b}{n}\cdot\frac{a}{n-1},$$

whatever the distinct positions r and s are. Since each X_k takes only the values 0 and 1, $\mathrm{E}(X_k) = \Pr(X_k = 1)$, and similarly $\mathrm{E}(X_iX_j) = \Pr(X_i = 1 \text{ and } X_j = 1)$.

Part (i)

For $i = 1$ we need the first letter to be A, so

$$\mathrm{E}(X_1) = \frac{a}{n}.$$

For $2 \le i \le n$ we need a B in position $i-1$ and an A in position i, so

$$\mathrm{E}(X_i) = \frac{b}{n}\cdot\frac{a}{n-1} = \frac{ab}{n(n-1)}.$$

By linearity of expectation, summing over the $n-1$ indices $i \ge 2$,

$$\mathrm{E}(S) = \frac{a}{n} + (n-1)\cdot\frac{ab}{n(n-1)} = \frac{a}{n} + \frac{ab}{n} = \frac{a(b+1)}{n},$$

as required.

Part (ii)

(a) Let $j \geq 3$. Then $X_1X_j = 1$ exactly when position 1 holds A, position $j-1$ holds B and position j holds A. Since $j-1 \geq 2$, these are three distinct positions, holding two As and one B. Filling them in turn,

$$\mathrm{E}(X_1X_j) = \frac{a}{n} \cdot \frac{b}{n-1} \cdot \frac{a-1}{n-2} = \frac{a(a-1)b}{n(n-1)(n-2)}.$$

(b) Now take $2 \leq i$ and $j \geq i+2$. Then $X_iX_j = 1$ exactly when positions $i-1$, i, $j-1$, j hold B, A, B, A respectively. Since $j-1 \geq i+1 > i$, these four positions are distinct, and they hold two As and two Bs, so

$$\mathrm{E}(X_iX_j) = \frac{b}{n} \cdot \frac{a}{n-1} \cdot \frac{b-1}{n-2} \cdot \frac{a-1}{n-3} = \frac{a(a-1)b(b-1)}{n(n-1)(n-2)(n-3)}.$$

(Here $n \geq 4$ because $a, b \geq 2$, so nothing vanishes in the denominator.) This is the same for every pair in the double sum, so we only need to count the pairs. For a given i, j runs from $i+2$ to n, which is $n-i-1$ values; as i runs from 2 to $n-2$ this takes the values $n-3$, $n-4$, …, 1. So the number of pairs is

$$1 + 2 + \cdots + (n-3) = \tfrac{1}{2}(n-3)(n-2),$$

and

$$\sum_{i=2}^{n-2} \left(\sum_{j=i+2}^{n} \mathrm{E}(X_iX_j) \right) = \frac{(n-2)(n-3)}{2} \cdot \frac{a(a-1)b(b-1)}{n(n-1)(n-2)(n-3)} = \frac{a(a-1)b(b-1)}{2n(n-1)}.$$

(c) We use $\mathrm{Var}(S) = \mathrm{E}(S^2) - (\mathrm{E}(S))^2$ and expand

$$S^2 = \sum_{i=1}^{n} X_i^2 + 2\sum_{i<j} X_iX_j.$$

Since X_i is 0 or 1, $X_i^2 = X_i$, so the first sum has expectation $\mathrm{E}(S)$.

Ask yourself What about the pairs with $j = i+1$, which part (b) left out?

They contribute nothing: $X_iX_{i+1} = 0$ always. For $i = 1$, $X_1 = 1$ needs an A in position 1 but $X_2 = 1$ needs a B there. For $i \geq 2$, $X_i = 1$ needs an A in position i but $X_{i+1} = 1$ needs a B there. (Two blocks of As can't start in neighbouring positions.)

So the pairs $i < j$ that matter are those with $j \geq i+2$. Those with $i = 1$ have $j = 3, \ldots, n$, which is $n-2$ pairs, each given by part (a); those with $i \geq 2$ are exactly the double sum in part (b). Hence

$$\begin{aligned}\sum_{i<j} \mathrm{E}(X_iX_j) &= (n-2) \cdot \frac{a(a-1)b}{n(n-1)(n-2)} + \frac{a(a-1)b(b-1)}{2n(n-1)} \\ &= \frac{a(a-1)b}{n(n-1)}\left(1 + \frac{b-1}{2}\right) = \frac{a(a-1)b(b+1)}{2n(n-1)},\end{aligned}$$

and therefore

$$\mathrm{E}(S^2) = \frac{a(b+1)}{n} + \frac{a(a-1)b(b+1)}{n(n-1)}.$$

Subtracting $(\mathrm{E}(S))^2 = a^2(b+1)^2/n^2$ and taking out the common factor $\dfrac{a(b+1)}{n^2(n-1)}$,

$$\mathrm{Var}(S) = \frac{a(b+1)}{n^2(n-1)}\Big[n(n-1) + n(a-1)b - a(b+1)(n-1)\Big].$$

Expanding the bracket, the terms nab and $-nab$ cancel, leaving

$$\begin{aligned} n^2 - n - nb - na + ab + a &= n(n-a-b) - n + ab + a \\ &= -n + ab + a \qquad \text{since } n = a+b \\ &= ab - b = b(a-1). \end{aligned}$$

So

$$\mathrm{Var}(S) = \frac{a(b+1)}{n^2(n-1)} \cdot b(a-1) = \frac{a(a-1)b(b+1)}{n^2(n-1)},$$

as required.

Checks

Take $a = 3$, $b = 2$, so $n = 5$ and there are 10 orderings. Counting blocks of As, the orderings $AAABB$, $BAAAB$, $BBAAA$ give $S = 1$; $ABABA$ gives $S = 3$; the other six give $S = 2$. So $\mathrm{E}(S) = \frac{18}{10} = \frac{9}{5}$ and $\mathrm{E}(S^2) = \frac{36}{10}$, giving $\mathrm{Var}(S) = \frac{18}{5} - \frac{81}{25} = \frac{9}{25}$. The formulae give $\frac{3\cdot3}{5} = \frac{9}{5}$ and $\frac{3\cdot2\cdot2\cdot3}{25\cdot4} = \frac{9}{25}$. A computer enumeration confirmed every formula in the question exactly for all $2 \le a, b \le 6$.

The formulae also behave sensibly outside the stated range. With $a = 2$, $b = 1$ the orderings AAB, ABA, BAA give $S = 1, 2, 1$, so $\mathrm{E}(S) = \frac{4}{3}$ and $\mathrm{Var}(S) = 2 - \frac{16}{9} = \frac{2}{9}$, matching $\frac{2\cdot2}{3}$ and $\frac{2\cdot1\cdot1\cdot2}{9\cdot2}$. And if $a = 1$ the variance formula gives 0, as it should, since then there is always exactly one block.

Common slips

- Treating X_1 like the others: there is no position 0, so X_1 needs only one letter and $\mathrm{E}(X_1) = a/n$, not $ab/(n(n-1))$.
- Forgetting that X_i and X_{i+1} can't both be 1, or computing $\mathrm{E}(X_iX_{i+1})$ with a non-zero formula.
- Miscounting the pairs in (b): there are $\frac{1}{2}(n-2)(n-3)$, not $\frac{1}{2}n(n-1)$.

Beyond STEP

Writing a count as a sum of indicator variables, then getting the mean by linearity and the variance from pairwise products, is one of the most useful techniques in

undergraduate probability; it is how one analyses runs, matchings, fixed points of random permutations and random graphs. Here S is essentially the statistic of the Wald–Wolfowitz runs test, used to detect non-randomness in a sequence. Blitzstein and Hwang, *Introduction to Probability*, has many examples of the indicator method.

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